

Theory of Automata

Assignment No: 1

Solution key

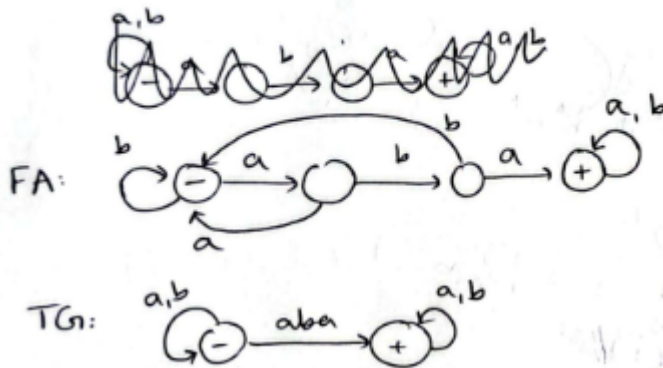
All language are defined over $\Sigma = \{a, b\}$

Question No:1

For the following languages

- Write an RE.
 - Design an FA, and
 - Design a TG. (only if the FA in part b can be expressed in a compressed way using a TG).
- i. All strings that contain at least one **aba** as a substring.

RE: $(a+b)^* (aba) (a+b)^*$

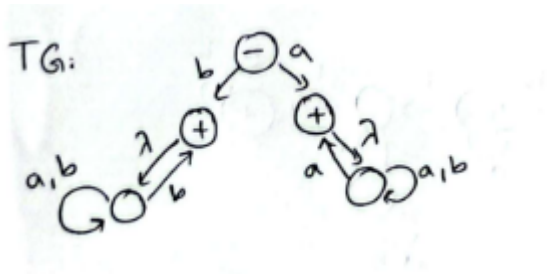
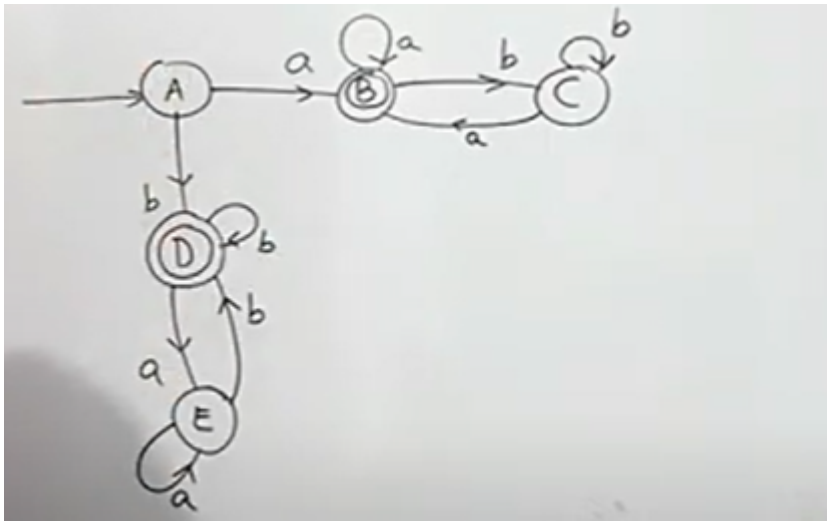


- ii. All strings that start and end with the same symbol.

RE:

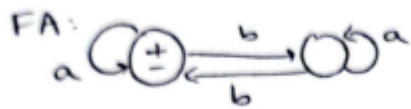
R.E: $((a+b)+a(a+b)^*a+b(a+b)^*b)^*$

FA:



iii. All strings in which the total number of **bs** is even.

$$a^*(ba^*ba^*)^*$$

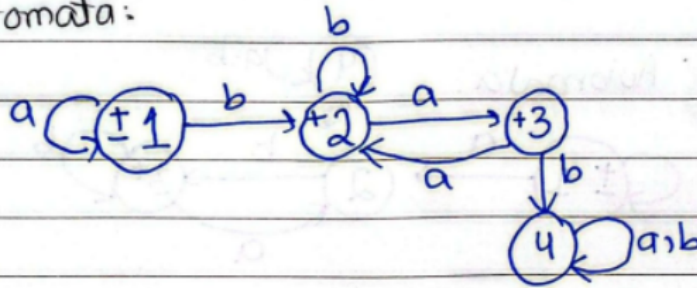


TG same as FA

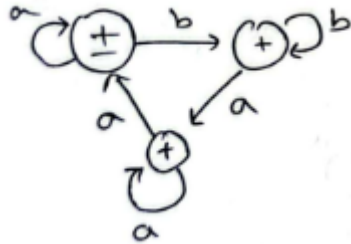
iv. All strings that do not contain the substring **bab**.

RE: $a^*(b^*aaa^*)^*b^*a^*$

Finite Automata:

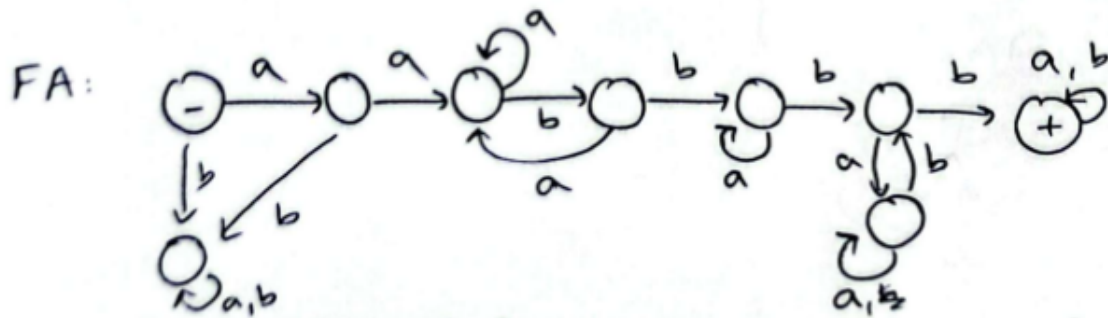


TG:

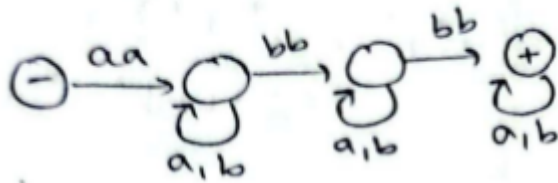


v. All strings that begin with aa and contain at least two bbs .

v. RE: $aa(a+b)^*bb(a+b)^*bb(a+b)^*$

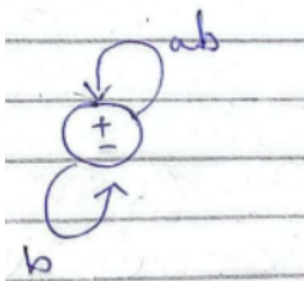
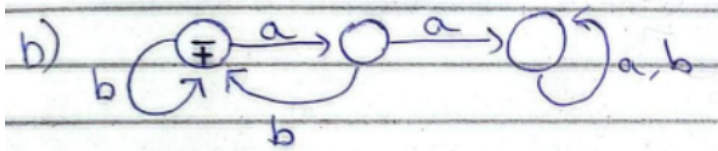


TG:



vi. All strings in which every **a** is immediately followed by at least one b

$(ab + b)^*$

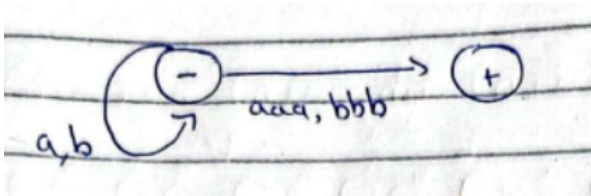
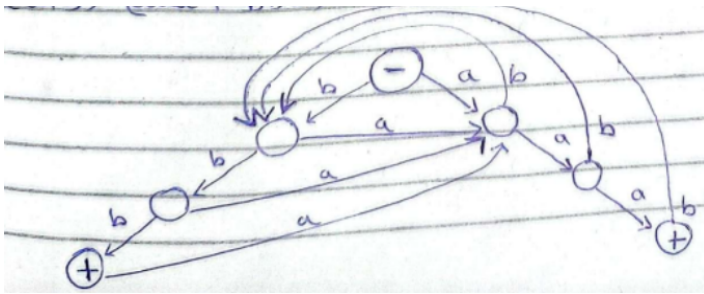


vii. All strings that contain an equal number of **a**s and **b**s .

Not possible as not included in Regular Language

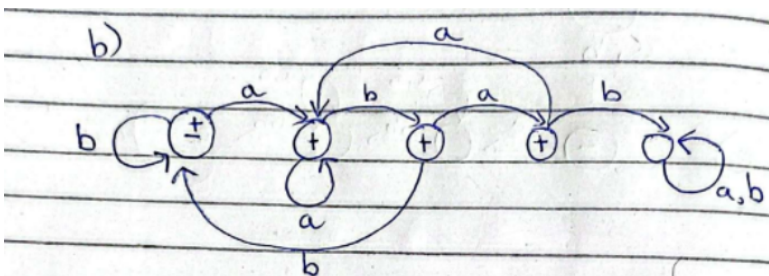
viii. All strings ending with either **aaa** or **bbb**.

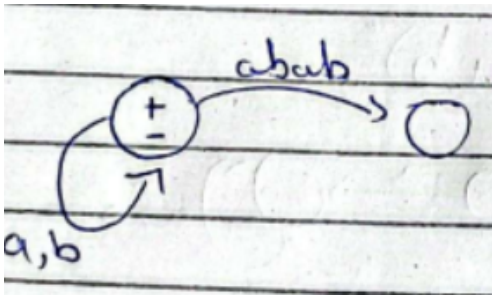
$(a + b)^* (aaa + bbb) \rightarrow RE$



ix. All strings that never contain **abab** as a substring.

$$b^*a^*b^*a^* + b^*a^*(bb^*a^*ab)^* \\ + (b^*abb^*ab)^*$$

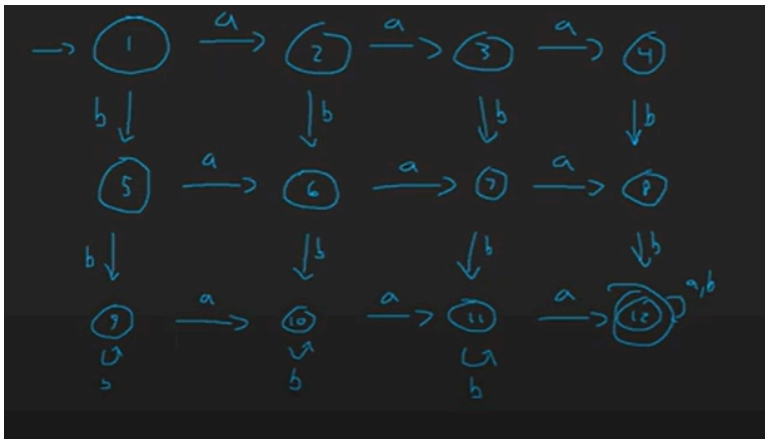




x. All strings with at least three **a**s and at least two **b**s.

RE: graded on the basis of the number of combinations you have written.

FA:



TG: same as FA

Question No:2

Describe (in English) the languages associated with the following regular expressions.

i. $(a+b)^*aa(a+b)^*$

A language of strings that contain substring "aa"

ii. $(ab+ba)^*$

The language is the set of all even-length strings over $\{a, b\}$ where every pair of

consecutive symbols is either "ab" or "ba". In other words, any string made by repeatedly concatenating "ab" or "ba" (including the empty string).

iii. $a^*b^*a^*$

The language of $a^*b^*a^*$ is all strings over $\{a,b\}$ where any b's (if present) appear in a single uninterrupted group, possibly preceded and/or followed by any number of a's.

iv. $(a+bb)^*$

Set of all strings over $\{a,b\}$ in which every b occurs as part of a doubled pair "bb", with any number of a's anywhere, including the empty string.

v. $b^*a(b^*ab^*a^*)^*$

This regular expression represents all strings over $\{a, b\}$ that contain at least one a. After first a, the string alternates between runs of bs and as with each cycle containing at least one a.

Question No:3

Show that

I. $((a+bb)^*aa)^*$ is equivalent to $\Lambda + (a+bb)^*aa$ in a sense they define the same language.

The handwritten proof is written on lined paper and shows the following steps:

- $\Rightarrow ((a+bb)^*aa)^*$: It makes strings with either start with a or bb.
- the strings will end with aa
- It can also have λ .
- $\Rightarrow \lambda + (a+bb)^*aa$: It also makes the strings that either start with a or bb.
- The strings end with aa.
- It can also have λ as shown.
- Thus, $((a+bb)^*aa)^* = \lambda + (a+bb)^*aa$

II. $a(ba + a)^*b$ is equivalent to $aa^*b(aa^*b)^*$ In a sense they define the same language.

$\Rightarrow a(ba + a)^*b$: It generates strings that start with a and end with b .

- It contains all combinations of ab , a and ba but no consecutive b 's.

$\Rightarrow aa^*b(aa^*b)^*$: It also generates strings that start with a and end with b .

- It also contains all combinations of a and b except consecutive b 's.

Thus, $a(ba + a)^*b = aa^*b(aa^*b)^*$

Question No: 4

I. Provide a recursive definition of language having all strings with length multiple of 2.

Step 1:

$a = \lambda, aa, bb, ab, ba$ are in language L .

Step 2: If x is in L , then xx is also in L .

Step 3: No strings except those constructed above are allowed to be in L .

II. Provide recursive definition of odd palindrome.

b-	Step 1: a, b is in odd palindrome	ii
	Step 2: If x is in odd palindrome, then axa and bxb are also in odd palindrome.	
	Step 3: No strings except those constructed above are allowed to be in odd palindrome.	iii

III. Give a recursive definition of the language ODDSTRING, i.e. of all words of odd length.

- 3)
- 1) a, b are in ODDSTRING
 - 2) x_1ax_2 and x_1bx_2 are in ODDSTRING if $x_1, x_2 \in \text{ODDSTRING}$ or x_1 and x_2 are both λ
 - 3) No strings except those constructed above belong to the language.

Instructions:

Submit assignment in hard form. Plagiarism gets minimum zero and maximum of F in course. No extension in deadline shall be given.